Random Forest

Katie Sylvia

GENERAL ASSEMBLY

Agenda

- Define how and why decision trees can be improved using bagging and random forests.
- Build random forest models for classification and regression.
- Explain how to extract the most important predictors in a random forest model.

Remind me...

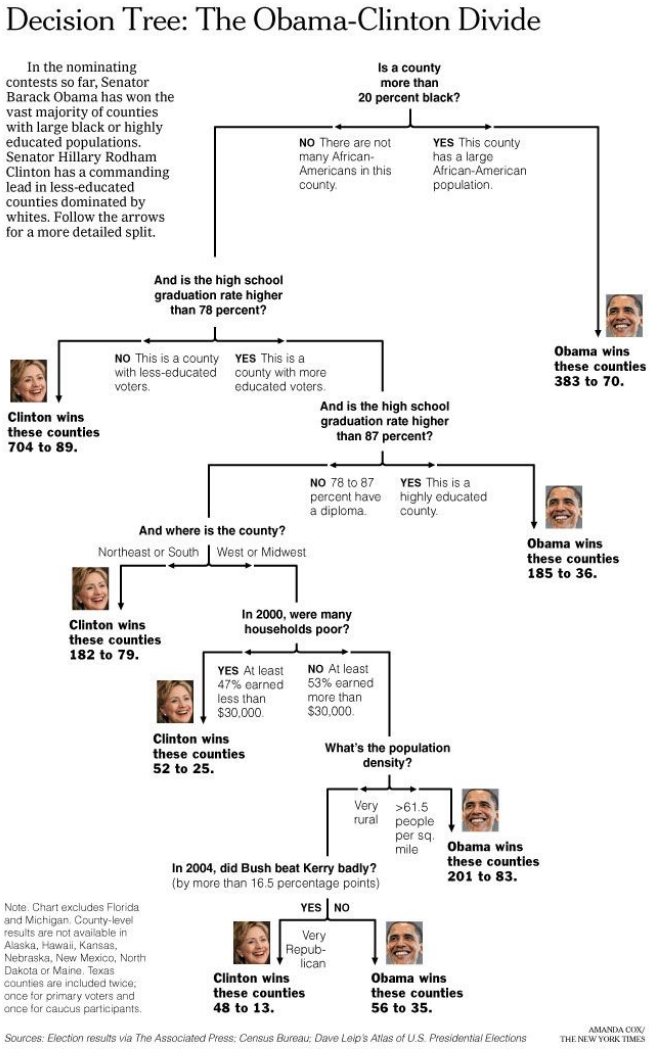
In your own words, what does a decision tree do?



This is an old graphic from the New York Times.

It explains how to predict whether Barack Obama or Hillary Clinton will win the Democratic primary in a particular county in 2008.

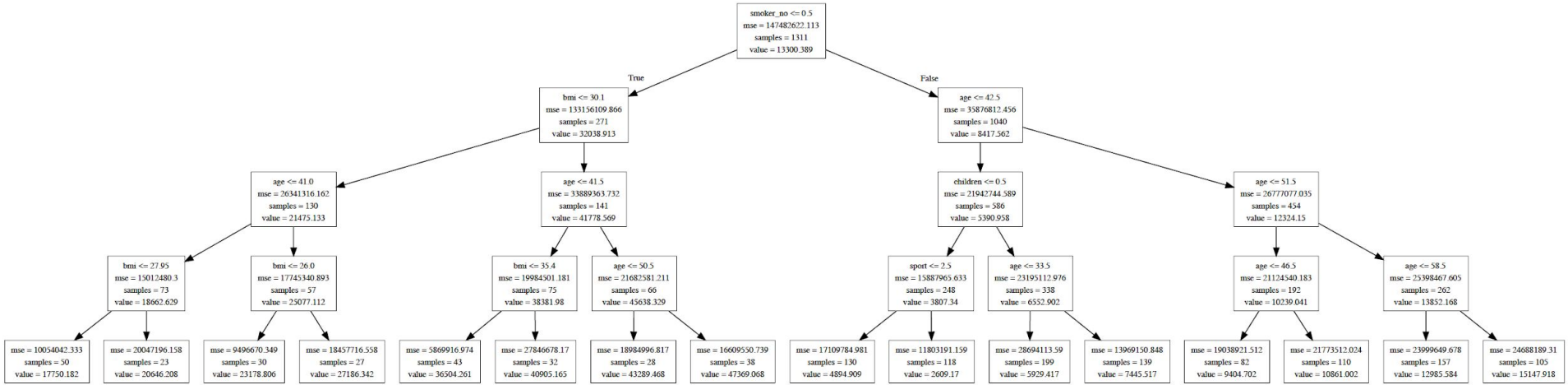
At the same time, it informs which features might be most important to predict an Obama or Clinton win.



Why Decision Trees?

- They can be applied to both regression and classification problems.
- They are easy to explain to others (interpretable). - They can be specified as a series of rules, and more closely approximate human decision-making than other models.
- They are very popular among data scientists.
- They are the basis for more sophisticated models.

White box, you say?



Why NOT Decision Trees?

- Overfitting - Decision-tree learners can create over-complex trees that do not generalize well from the training data.
- Decision tree learners create biased trees if some classes dominate. It is therefore recommended to balance the dataset prior to fitting with the decision tree.
- They don't tend to work well with very small data sets.

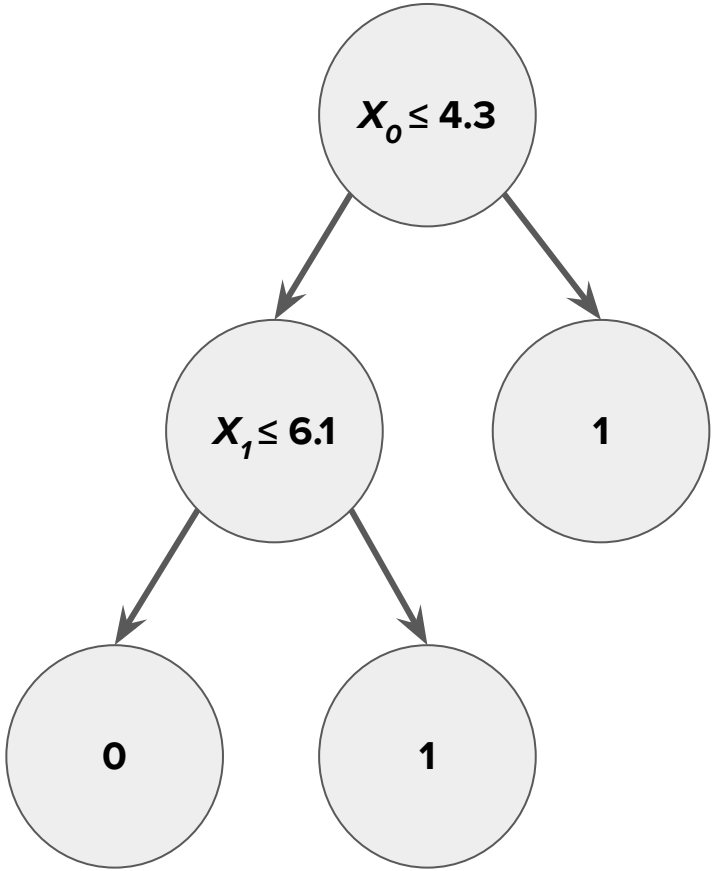
Credit Where Credit is Due

The next few slides in this lesson was developed with help from this source:

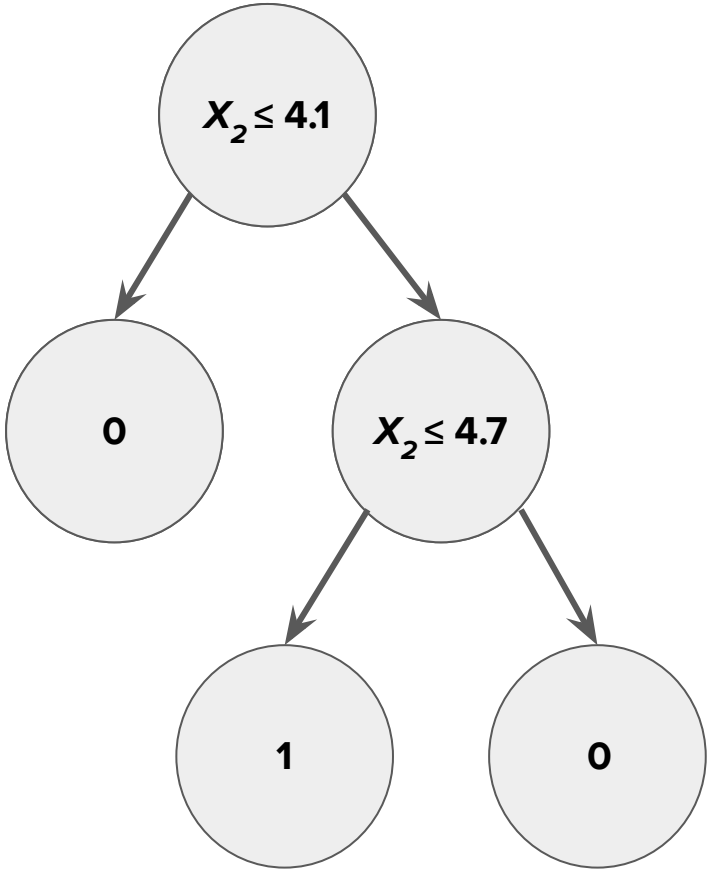
- Normalized Nerd's video called 'Random Forest Algorithm Clearly Explained!' found here.



id	X_0	X_1	X_2	X_3	X_4	y
0	4.3	4.9	4.1	4.7	5.5	0
1	3.9	6.1	5.9	5.5	5.9	0
2	2.7	4.8	4.1	5.0	5.6	0
3	6.6	4.4	4.5	3.9	5.9	1
4	6.5	2.9	4.7	4.6	6.1	1
5	2.7	6.7	4.2	5.3	4.8	1



id	X_0	X_1	X_2	X_3	X_4	y
0	4.3	4.9	4.1	4.7	5.5	0
1	6.1	4.1	5.9	5.5	5.9	0
2	2.7	4.8	4.1	5.0	5.6	0
3	6.6	4.4	4.5	3.9	5.9	1
4	6.5	2.9	4.7	4.6	6.1	1
5	2.7	6.7	4.2	5.3	4.8	1




We'll have another tree here.
No use stopping when we're having
so much fun here with these trees!

id	X_0	X_1	X_2	X_3	X_4	y
0	4.3	4.9	4.1	4.7	5.5	0
1	3.9	6.1	5.9	5.5	5.9	0
2	2.7	4.8	4.1	5.0	5.6	0
3	6.6	4.4	4.5	3.9	5.9	1
4	6.5	2.9	4.7	4.6	6.1	1
5	2.7	6.7	4.2	5.3	4.8	1

id	id	id	id
2	2	4	3
0	1	1	3
2	3	3	2
4	1	0	5
5	4	0	1
5	4	2	2

Bootstrapped Datasets

id	X_0	X_1	X_2	X_3	X_4	y
0	4.3	4.9	4.1	4.7	5.5	0
1	3.9	6.1	5.9	5.5	5.9	0
2	2.7	4.8	4.1	5.0	5.6	0
3	6.6	4.4	4.5	3.9	5.9	1
4	6.5	2.9	4.7	4.6	6.1	1
5	2.7	6.7	4.2	5.3	4.8	1

id
2
0
2
4
5
5

X_0, X_1

id
2
1
3
1
4
4

X_2, X_3

id
4
1
3
0
0
2

X_2, X_4

id
3
3
2
5
1
2

X_1, X_3

For a problem with p features, it is typical to use:

Classification - $\sqrt{p}$ (rounded down) features
Regression - $p/3$ (rounded down) features

In our example, we have 5 features.

$\sqrt{5} \simeq 2.236... \rightarrow$ 2 (rounded down) features

id	X_0	X_1	X_2	X_3	X_4	y
0	4.3	4.9	4.1	4.7	5.5	0
1	3.9	6.1	5.9	5.5	5.9	0
2	2.7	4.8	4.1	5.0	5.6	0
3	6.6	4.4	4.5	3.9	5.9	1
4	6.5	2.9	4.7	4.6	6.1	1
5	2.7	6.7	4.2	5.3	4.8	1

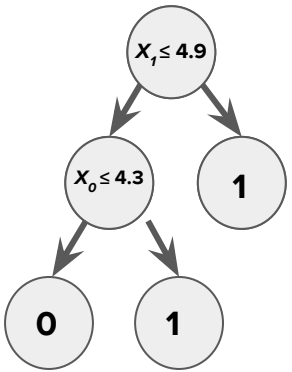
id
2
0
2
4
5
5

id
2
1
3
1
4
4

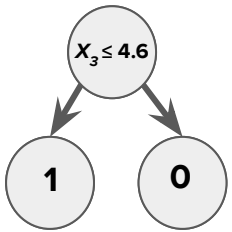
id
4
1
3
0
0
2

id
3
3
2
5
1
2

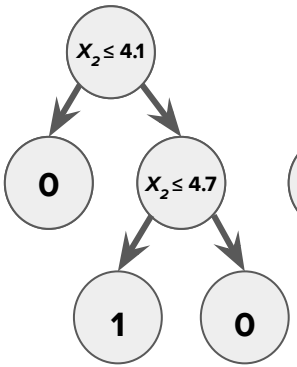
X_0, X_1



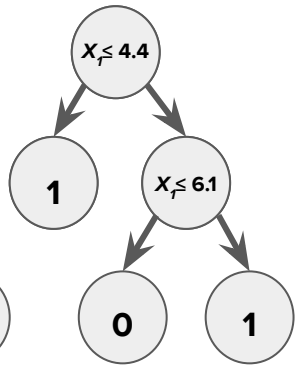
X_2, X_3



X_2, X_4



X_1, X_3



id	X_0	X_1	X_2	X_3	X_4	y
0	4.3	4.9	4.1	4.7	5.5	0
1	3.9	6.1	5.9	5.5	5.9	0
2	2.7	4.8	4.1	5.0	5.6	0
3	6.6	4.4	4.5	3.9	5.9	1
4	6.5	2.9	4.7	4.6	6.1	1
5	2.7	6.7	4.2	5.3	4.8	1

2.8	6.2	4.3	5.3	5.5

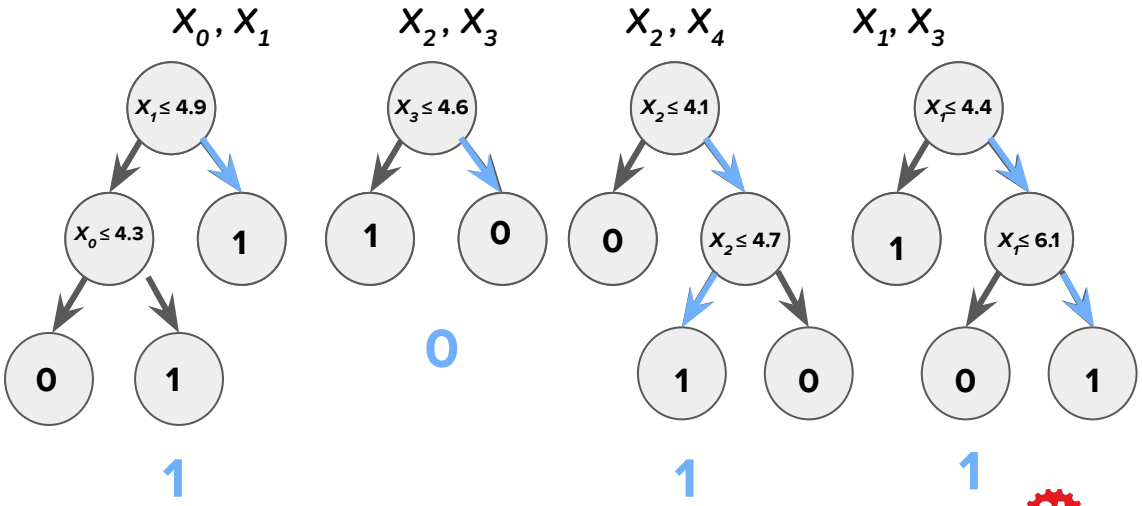
Bootstrap + Aggregation = Bagging

id
2
0
2
4
5
5

id
2
1
3
1
4
4

id
4
1
3
0
0
2

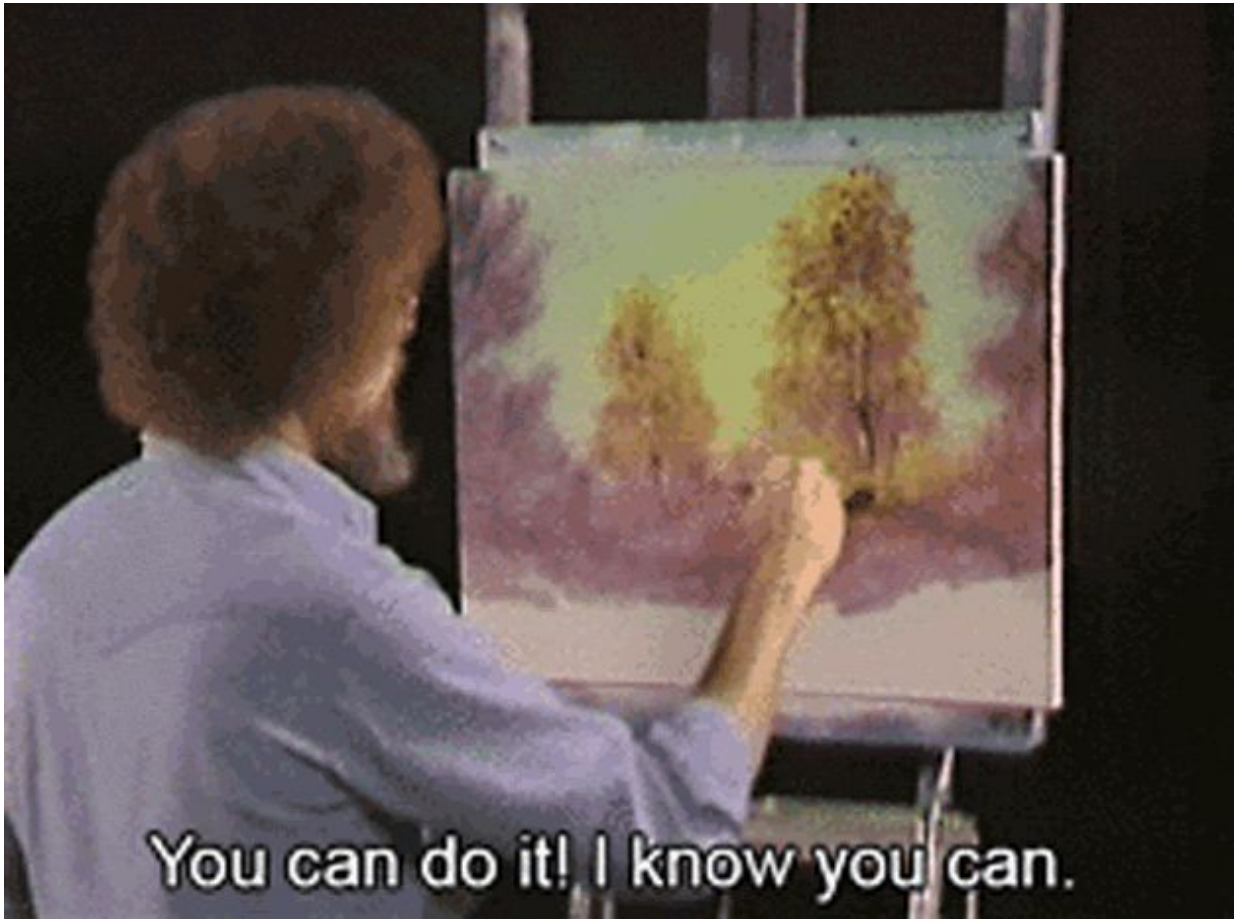
id
3
3
2
5
1
2



So...

How are we feeling about Random Forest?




You can do it! I know you can.

Extra Trees

GENERAL ASSEMBLY

Extra Trees

- SIMILARITIES:
  - Like Random Forests, these are trained using the random subspace method (sampling of features).
- DIFFERENCES:
  - They are trained on the entire dataset instead of bootstrapped samples! So Extra Trees samples WITHOUT replacement.
  - A layer of randomness is introduced in the way the nodes are split. Instead of computing the locally optimal feature/split combination (based on, e.g., information gain or the Gini impurity) for each feature under consideration, a random value is selected for the split. This leads to more diversified tree.
  - This further reduces the variance, but causes an increase in bias.

Extra Trees

- On toy datasets, the following conclusions could be reached :
  - Extra trees seem to keep a higher performance in presence of noisy features
  - When all the variables are relevant, both methods seem to achieve similar performance.
- Can be best to build an ExtraTrees model and a Random Forest model, then compare their performance!
- That’s exactly what we’ll do.

Let's try them out!




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